



Engineering Precision
That Feeds The World.

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COMPANY PROFILE

Complete Turnkey Solutions For Fish Feed, Poultry Feed
& Biomass Production Lines.

TABLE OF CONTENT

Table of Contents

Company Profile	04-05
About Company	05
CEO Message	06-07
Livestock & Poultry Feed Production Line	08-09
Fish Feed Production Line	10-11
Biomass Feed Production Line	12-13
Hammer Mill	14-15
Mixer Units	16-17
Extruder	17-18
Feed Pellet Mill	19-20
Cooling & Dryer Equipment	21-22
Crumbler & Screener	23-24
Cleaner & Packing Scales	25-26
Conveyor & Ancillary Equipment	27-30
Preformed Units & Spare Parts	31-32
Projects & Client Highlights	33-38

ABOUT COMPANY

Founded in 1994 in Lahore, Pakistan, SEMZONE Engineering began as a small manufacturing shop and has since grown into one of the country's leading engineering enterprises. Starting with poultry feed machinery, we expanded into advanced sectors including fertilizer and pet food production equipment.

Today, SEMZONE delivers design, engineering, manufacturing, installation, and after-sales services across Pakistan and multiple Asian markets. With decades of experience, we have established a reputation for innovation, reliability, and sustainable solutions.

Our Expertise

We specialize in providing high-performance production lines and machinery, including:

- Animal Feed Production Plants – 2 to 60 tons/hour
- Aqua Feed Production Plants – 1 to 20 tons/hour
- Biomass Production Plants – 1 to 20 tons/hour
- Single Super Phosphate (SSP) Plants – 50 to 200 tons/day
- Organic Compost Fertilizer Units – 50 to 100 tons/day
- Bio Organic Phosphate (BOP) Plants – 50 to 200 tons/day
- Storage Tanks – 1 to 100 tons for oil & water

Our facilities also cover NPK production lines, compound fertilizer machinery, compost units, and biomass pellet systems. Each solution is engineered for efficiency, safety, and environmental responsibility.



Why SEMZONE

- Trusted by customers for quality and precision.
- Extensive track record in fertilizer, poultry feed, and aqua feed machinery.
- Focus on eco-friendly and reliable engineering solutions.
- Commitment to long-term value and customer satisfaction.

Our guiding principle: "Quality First, Service First, Price Reasonable."

Vision

To be Pakistan's leading engineering company, setting new benchmarks in innovation, sustainability, and excellence. We aim to deliver world-class projects that foster growth, empower communities, and build a legacy of trust and reliability.



MISSION

At Samzon Engineering, our mission is to be at the forefront of the industrial machinery and feed solutions industry, delivering high-performance products and services that empower our customers to achieve superior efficiency, productivity, and sustainability. We understand the evolving needs of agriculture, livestock, and feed industries, and our goal is to provide innovative solutions that meet these demands with unmatched quality, reliability, and precision.

Our commitment extends beyond providing state-of-the-art machinery and feed solutions. We aim to build lasting relationships with our customers by offering expert guidance, personalized service, and a deep understanding of the challenges they face. Through our extensive experience, cutting-edge technology, and dedicated research and development efforts, we strive to deliver products that not only increase operational efficiency but also contribute to the overall success and growth of the businesses we serve.

We believe in continuous improvement and innovation, driving us to develop sustainable solutions that align with the global shift towards eco-friendly practices in agriculture and industry. By maintaining the highest standards of integrity, accountability, and customer satisfaction, Samzon Engineering is dedicated to becoming the trusted partner of choice for businesses looking to enhance their operations with reliable, advanced machinery and feed solutions. Together with our customers, we aim to build a brighter, more efficient future for industries across the globe.

CEO MESSAGE

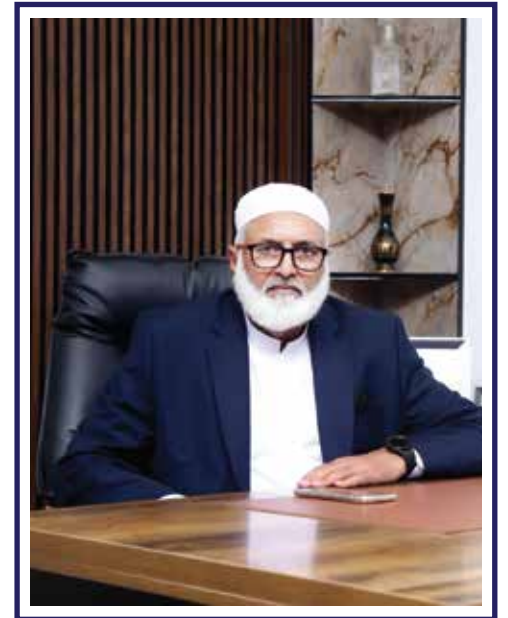
At Samzon Engineering, our journey has always been guided by a vision to innovate, create, and lead with integrity. From humble beginnings to becoming a trusted name in engineering solutions, our focus has remained the same: delivering excellence, reliability, and value to our clients.

We believe that true progress lies in combining advanced technology with sustainable practices. Every product we design and manufacture reflects our commitment to quality, safety, and efficiency. With a strong team of professionals and state-of-the-art facilities, we aim not only to meet expectations but to consistently exceed them.

As we look ahead, our mission remains clear: to strengthen our position as a leader in engineering solutions, expand our global footprint, and contribute to industries that are shaping a better and more sustainable future. We extend our sincere gratitude to our valued clients and partners for their continued trust and collaboration, which drive us to reach new heights of excellence.

With innovation at our core, we are dedicated to developing efficient, high-quality, and environmentally responsible solutions that empower progress across diverse sectors. Backed by a passionate team and a strong commitment to quality, we continue to evolve — setting new benchmarks, building lasting partnerships, and paving the way toward a smarter and stronger tomorrow.

At Samzon Engineering, we don't just build systems — we build trust, progress, and a legacy of innovation that stands the test of time. Our future lies in pushing boundaries, embracing change, and inspiring the next generation of engineers to think boldly and create fearlessly. Together, we are not just shaping the future of engineering — we are engineering the future itself.



Together, we engineer progress.

Shoukat Ali
Chief Executive Officer
Samzon Engineering

Poultry Feed

Flow Chart Process

Raw Material



Feed Grinding

Raw materials (maize, soybean meal, wheat) are ground to 500–700 µm particle size using a hammer mill for better mixing and pellet quality.



Feed Cleaner

Removes dust, husk, and magnetic impurities to protect machinery and ensure feed purity.



Feed Mixing

Macro, micro, and liquid ingredients blended for uniform distribution (CV ≤ 5%) within 90–120 seconds.



Pellet Mill

Conditioned with steam at 80–85°C and 15–17% moisture, then compressed into Ø 3–4 mm pellets for durability and improved intake.



Cooler

Pellets cooled to ambient +5°C; final moisture reduced to 11–12% for stability and shelf life.3. Mixer



Crumbler

Pellets crushed into 0.8–1.8 mm crumbles for chick starter feed; maintains uniform size and structure.



Sieve

Screens fines and oversized particles; ensures consistent pellet and crumble size before packing.



Packaging

Feed packed in 25/50 kg bags or dispatched in bulk; labeled for traceability and quality control.

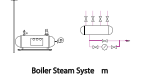
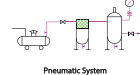
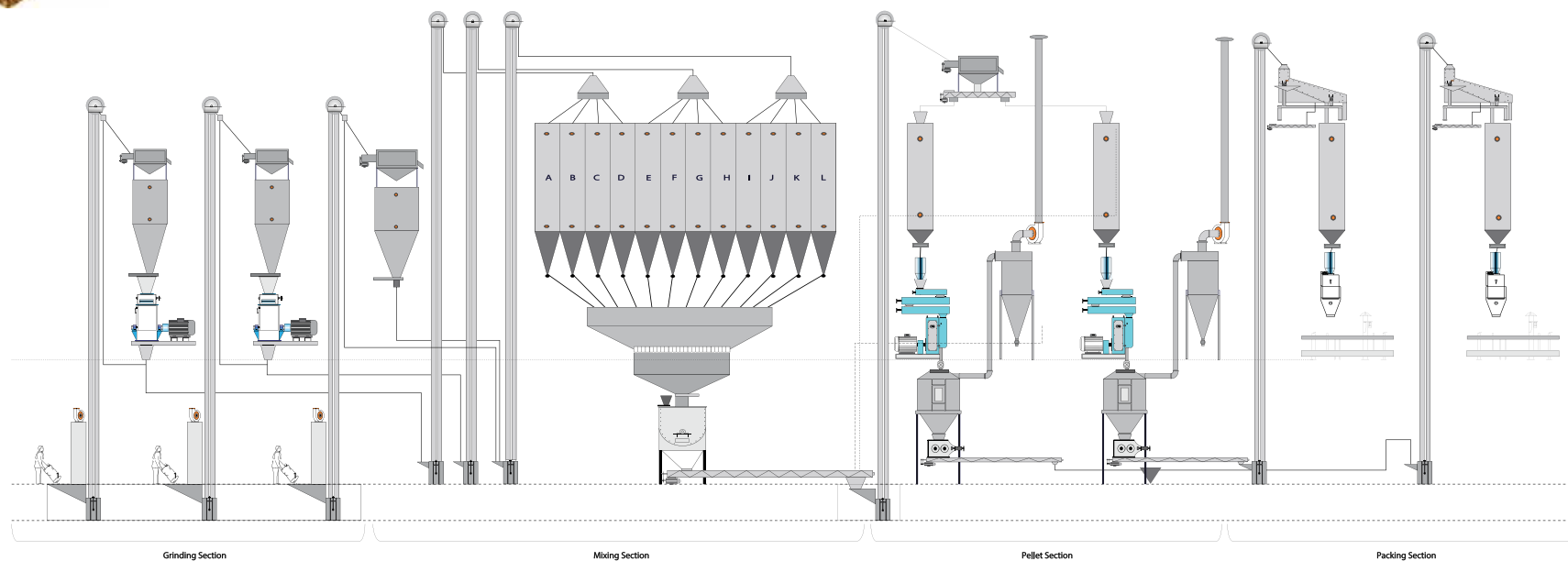


Poultry Feed



Samzon Engineering
livestock and poultry feed production
line

>10–80t/h chicken feed
production line
> 10–80t/h duck and goose feed
production line



Boiler

Samzon Engineering's industrial steam boilers are designed specifically for feed mills, ensuring stable pressure, high steam output, and low fuel consumption. Built with robust construction and advanced safety systems, they deliver reliable performance even under heavy daily load.



Electronic Panel

This control panel centralizes operation of key feed mill units such as hammer mill, mixer, screw conveyors, elevators, pellet mill, conditioner, blower, cooler, and packing machines. The system is engineered for continuous operation, high reliability, and precise control over every process stage.

FISH Feed

Flow Chart Process

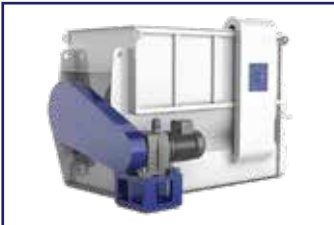
Raw Material



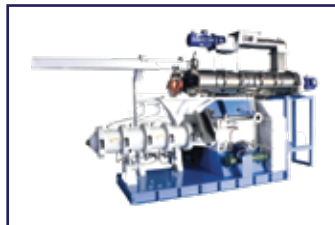
Pre-Grinding
Raw ingredients (fish meal, soybean meal, corn) are pre-ground to 300–500 µm for uniform texture and better extrusion performance.



Mixer
Dry and liquid ingredients blended evenly (CV ≤ 5%) to ensure balanced nutrition in every pellet.



Extrusion
Conditioned feed processed through an extruder at 100–130 °C with 18–20% moisture to form floating or sinking pellets with controlled density.



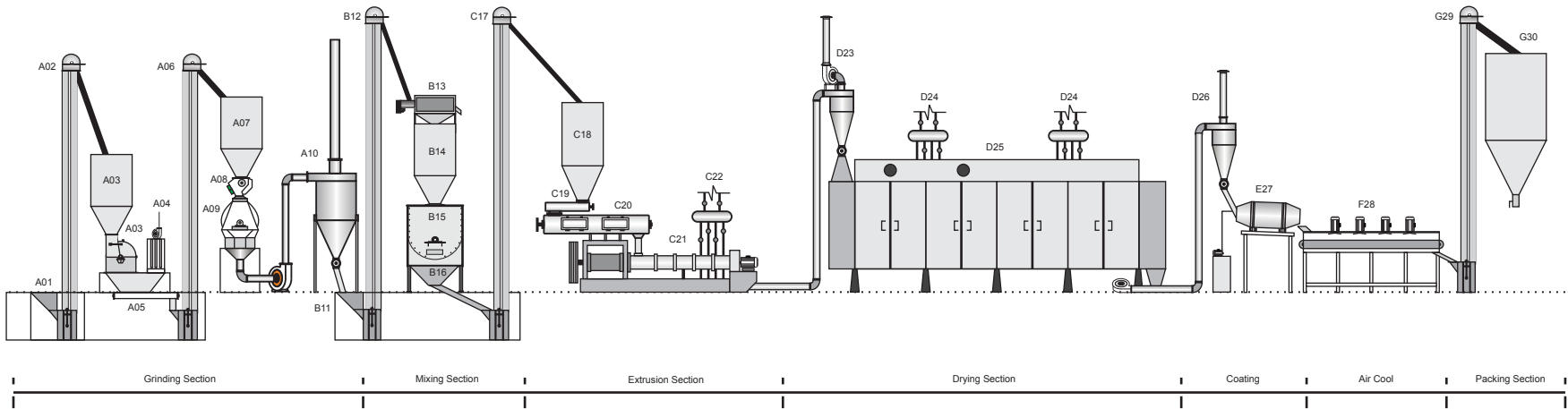
Steam Dryer
Hot air and steam dry pellets to 9–10% moisture, improving hardness and shelf life.



Samzon Engineering
Fish Feed Production Line

>2–20t/h extruded fish feed
production line

>2–20t/h pet food production line



Cooler (Oil Coating)
Pellets coated with oil, enzymes, or vitamins for energy enrichment and nutrient retention.



Fish Feed

Packing
Finished feed packed in 25 / 50 kg bags or bulk; labeled with batch number, pellet size, and product type.



Vibration
Fines and oversized pellets separated; only uniform pellets proceed for packing.



Cooling
Cooled to ambient + 5 °C to stabilize structure and prevent moisture condensation.



BIOMASS WOOD PELLET

Flow Chart Process

Raw Material



Chipper Mill
This heavy-duty chipper features a compact and well-balanced structure, offering high productivity, excellent chip quality, and easy operation.



Pre-Grinding
Raw ingredients (fish meal, soybean meal, corn) are pre-ground to 300-500 µm for uniform texture and better extrusion performance.



Screener
Removes dust, husk, and magnetic impurities to protect machinery and ensure feed purity.

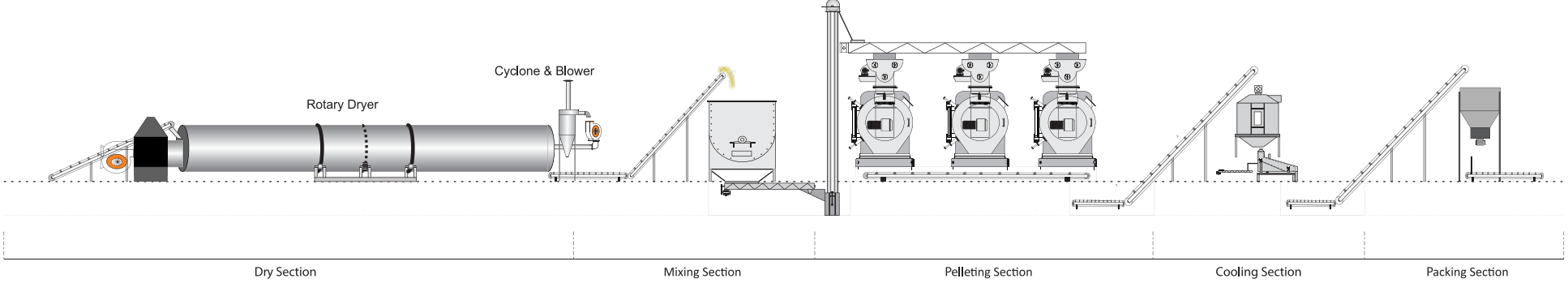
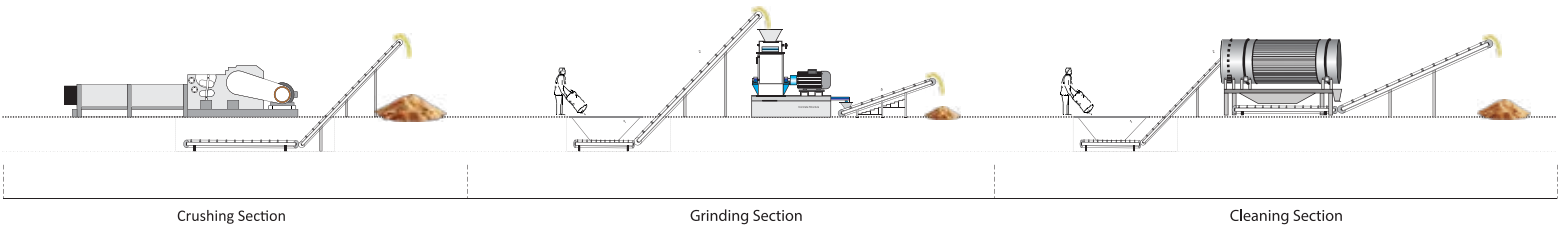


Dryer
Biomass materials (sawdust, rice husk, wood chips, straw) are dried to reduce moisture from 35-45% + 10-12%, ensuring efficient pelletizing and stable burning quality.



Samzon Engineering
Biomass Feed Production Line

2-20t/h wood pellet
production line



Mixer
Dried material is blended to achieve uniform moisture and particle distribution, improving pellet density and durability.



Pellet Mill
Material is compressed through a ring die under 80-100 °C to form 6-10 mm biomass pellets with high density and smooth surface.



Auto Packing
Finished biomass pellets are automatically weighed and packed into 15-50 kg bags or bulk. The system ensures accuracy, speed, and clean operation.



Vibrator
Removes dust, powder, and broken pellets; ensures consistent size and high-quality output before packing.



Cooler
Fresh pellets are cooled to ambient +5 °C, reducing temperature and moisture to ≤10% to prevent cracking and mold formation.



Manual Cattle Feed

Flow Chart Process

Raw Material



Feed Grinding

Raw materials (maize, soybean meal, wheat) are ground to 500-700 µm particle size using a hammer mill for better mixing and pellet quality.



Feed Cleaner

Removes dust, husk, and magnetic impurities to protect machinery and ensure feed purity.



Feed Mixing

Macro, micro, and liquid ingredients blended for uniform distribution (CV ≤ 5%) within 90-120 seconds.



Pellet Mill

Conditioned with steam at 80-85°C and 15-17% moisture, then compressed into Ø 3-4 mm pellets for durability and improved intake.



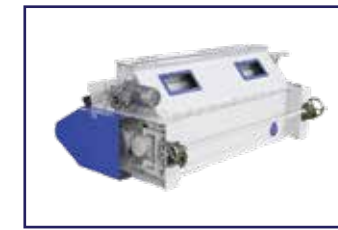
Cooler

Pellets cooled to ambient +5°C; final moisture reduced to 11-12% for stability and shelf life. 3. Mixer



Crumbler

Pellets crushed into 0.8-1.8 mm crumbles for chick starter feed; maintains uniform size and structure.



Sieve

Screens fines and oversized particles; ensures consistent pellet and crumble size before packing.

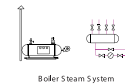
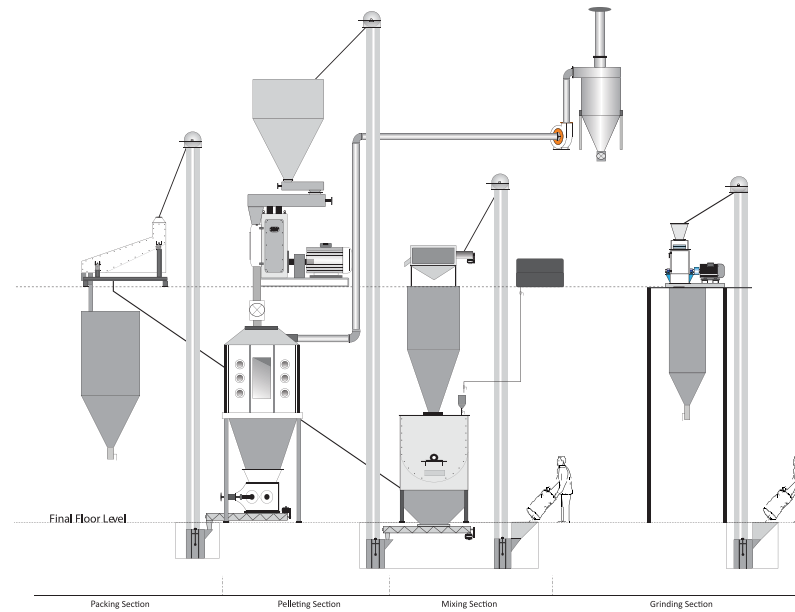


Packaging

Feed packed in 25/50 kg bags or dispatched in bulk; labeled for traceability and quality control.



Poultry Feed



Boiler

Semzon Engineering's industrial steam boilers are designed specifically for feed mills, ensuring stable pressure, high steam output, and low fuel consumption. Built with robust construction and advanced safety systems, they deliver reliable performance even under heavy daily load.



Electronic Panel

This control panel centralizes operation of key feed mill units such as hammer mill, mixer, screw conveyors, elevators, pellet mill, conditioner, blower, cooler, and packing machines. The system is engineered for continuous operation, high reliability, and precise control over every process stage.

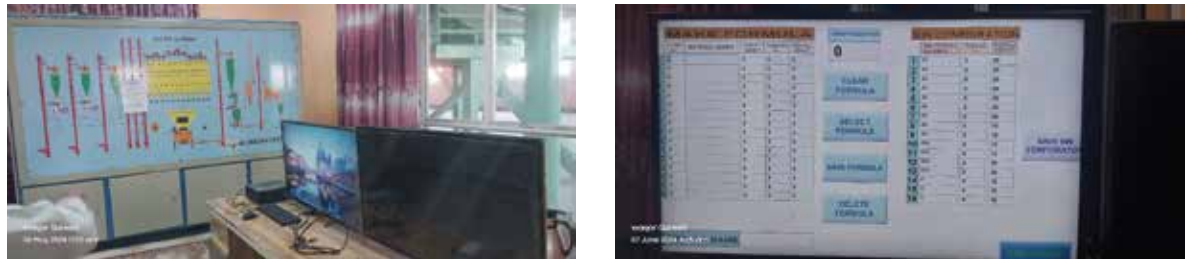
PROGRAMMABLE LOGIC CONTROLLER

A Programmable Logic Controller (PLC) is a powerful industrial automation unit designed to control, regulate, and monitor complex machinery with exceptional precision. Engineered to operate reliably in harsh industrial environments, a PLC ensures continuous performance even under conditions of vibration, dust, heat, and heavy workloads. It receives signals from sensors and input devices, processes them through a programmed logic sequence, and delivers accurate output commands to motors, actuators, and other equipment—making it the operational backbone of modern automated systems. With its modular architecture, the PLC allows easy expansion of input/output modules as production requirements grow, while its user-friendly interface supports multiple programming languages such as ladder logic, function block, and structured text.



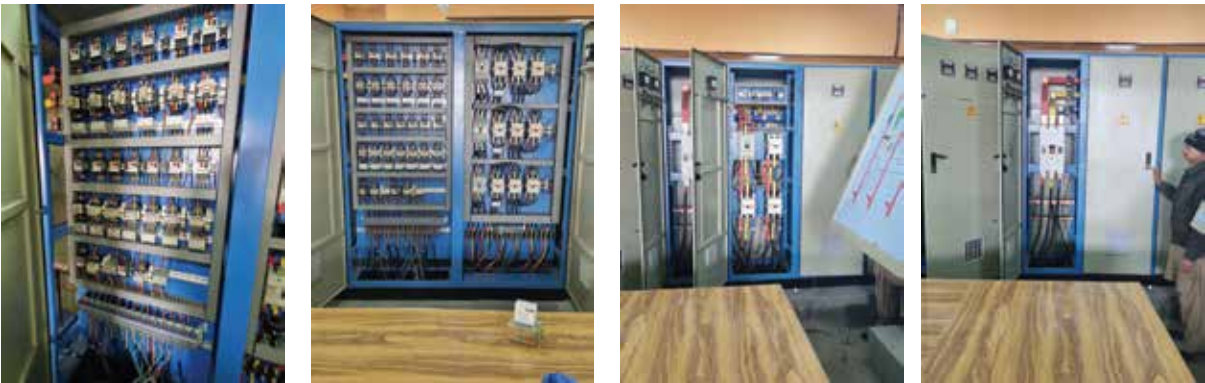
Highlights

- High-speed processing for smooth automation.
- Reliable performance in heat, dust & vibration.
- Modular I/O design for easy expansion.
- Simple programming using standard industrial languages.
- Real-time monitoring and fault detection.



ELECTRONIC CONTROL SYSTEM

An Electronic Control System is a highly efficient automation unit designed to manage, regulate, and optimize the performance of modern industrial machinery. Using advanced electronic components and intelligent control logic, the system monitors key parameters in real time and ensures every part of the machine operates with maximum precision and stability. It receives data from sensors, processes it through integrated circuits and microcontrollers, and delivers accurate output signals to motors, valves, actuators, and safety devices. This creates a smooth, fully synchronized workflow across the entire production line. Built for reliability, the Electronic Control System performs consistently even under demanding industrial conditions such as heat, vibration, and continuous 24/7 operation. Its modular design supports expansion and customization, allowing industries to add new functions or upgrade components as their automation needs grow.



FEATURES

- Precise digital control
- Real-time parameter monitoring
- Easy-to-use interface
- Modular and expandable design
- High reliability in tough conditions
- Smart safety alarms & interlocks
- Low power consumption
- Fast fault detection
- Smooth and stable machine output

With user-friendly operation, digital displays, fault indicators, and smart safety interlocks, it reduces down-time, minimizes human error, and improves overall machine performance. Whether used in feed mills, extrusion plants, conveyors, packaging units, weighing systems, or automated batching setups, the Electronic Control System delivers unmatched control accuracy, energy efficiency, and long-term operational durability—making it an essential part of any modern industrial setup aiming for high productivity and dependable results.

HAMMER MILL

This high efficiency hammer mill is more suitable for raw material fine grinding.



Model	SMSP965X400	SMSP965X500	SMSP965X600	SMSP965X800
Main Shaft Speed(rpm)	1480	1480	1480	1480
Power(kw)	55/75	90/110	132/160	200/220
Capacity(t/h)	5-10	8-12	15-20	20-35

Note: The above indices are valid when meeting the following conditions: corn as raw material; moisture no higher than 12%; density no lower than 0.72t/m³; hole diameter of screen - 3mm; hole-opening rate no less than 33%.

HAMMER MILL

High-performance grinding for feed & biomass industries.



Model	SMSP600X380	SMSP600X450	SMSP600X500	SMSP600X600
Main Shaft Speed(rpm)	1480	1480	1480	1480
Power(kw)	30/37	45/55	55/75	75/90/110
Capacity(t/h)	2-4	3-6	5-8	8-12

RIBBON MIXER

The Horizontal Corn Mixer Machine is a high-efficiency mixing solution designed for use in premix, poultry feed, livestock feed, and aquafeed production lines. Built with advanced engineering and durable materials, this mixer ensures uniform blending of powdered and granular raw materials, enhancing feed quality and consistency.

KEY FEATURES

- Specially designed discharger has big door-opening angle and quick discharging with little residue.
- Good tightness and no leakage. Pneumatic or Electric driven discharging is used with correct, stable move and good tightness.
- Applicable for compound feed plant, premix feed plant, food and flour plants.
- Gentle mixing, without deviation and good homogeneity.



Model	Capacity (kg/batch)	Power (kw)
SMHY0.6	250	5.5/11
SMHY1	500	11/15
SMHY2.5	1000	15/22
SMHY5	2000	37/45

PADDLE MIXER DOUBLE SHAFT

KEY FEATURES

- Can be equipped with compressed air cleaning system to reduce residual to the lowest level.
- Full-length door opening and discharging mechanism specially designed, with large opening Angle, rapid discharging and less residue. Good sealing, ensure no leakage, large range of loading capacity.
- Fast mixing speed, high uniformity, no dead Angle, short mixing cycle



Model	Effective Volume(m ³)	Mixing Capacity Per Batch (kg)	Mixing Homogeneity (Cv ≤ %)	Power (kw)
SMSJ0.5	0.5	250	5	5.5/7.5
SMSJ1	1	500	5	11/15
SMSJ2	2	1000	5	22
SMSJ4	4	2000	5	37/45
SMSJ6	6	3000	5	45/55

EXTRUDER

KEY FEATURES

- Effectively allocate reasonable shear forces in different areas of the extrusion chamber, and increase the yield under the premise of ensuring high quality.
- The use of dual-axis differential quality regulator to ensure full regulation and high efficiency.
- Equipped with international advanced standards, the equipment can improve the stability of the equipment and reduce the noise.



Model	Capacity(t/h)	Main Power(KW)	Feeder Power(KW)	Conditioner Power (kw)
SMHS128	0.8-1.2	45/55	1.5	3
SMHS135	1.5-2.0	55/75	1.5	4
SMHS165	3.0-5.0	132/160	1.5	4
SMHS185	4.0-5.0	160/185	1.5	5.5

EXTRUDER

Be exclusively used in extruding soybean, corn, bean pulp, rice bran and other raw material. It also can be used in beans pretreatment.

KEY FEATURES

- Shearing action of high temperature and high pressure, sterilizing detoxification effectively, change the antinutritional factors, reduce the degradation of nutrients, improve the digestibility of protein and starch.
- Horizontal force feeding structure, improve the production capacity of more than 20% effectively.
- The real multi-functional equipment. Can produce expanded soybean, expanded piglet feed and other feed and reduce cost only if replacing small amount of spare parts.



Model	Capacity (t/h)			Power of Motor
	Corn	Soybean meal	Soybeans	
SPHG2500	1.4-1.0	1.0-2.5	0.6-1.8	55/75
SPHG3200	1.0-1.8	3.0-4.0	1.5-2.5	90/110
SPHG5000	1.5-2.5	3.0-4.0	2.5-4.0	110/132

FEED PELLET MILL



Model	Main Power(KW)	Capacity t/h	Feeder Power(KW)	Conditioner Power(KW)	Die Diameter (mm)
SZLH350	55	2-5	0.7	3	350
SZLH400	75/90	3-8	1.1	4	400
SZLH420	110	5-12	1.1	5.5	420
SZLH508	132/160	8-15	1.5	7.5	508
SZLH578	160/185	10-20	1.5	7.5	558
SZLH678	200/250	15-28	2.2	11	678
SZLH768	250/280/315	20-35	2.2	11	768



pellet mill is mainly used for large sized feed mills producing kinds of granule feed which can reduce the cost. This pellet mill is more suitable for granule feed production of large-sized feed mills.

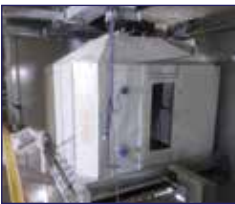
KEY FEATURES

- High power and large output, stable operation, easy operation and maintenance.
- Die diameter varies from 350mm to 768mm; the maximum capacity is up to 35t/h.
- Our poultry feed processing equipment with simple structure, strong adaptability, a small footprint, and low noise.

COOLING EQUIPMENT

KEY FEATURES

- Adopt counterflow cooling principle to make the pellet cooling fully and uniformly.
- Pellet temperature is not higher than room temperature + 3~5 °C after cooling.
- Slide valve discharging structure running smoothly.

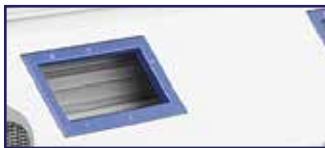


Model	SKLN2.5 (15×15)	SKLN4 (20×20)	SKLN6 (22×22)	SKLN8 (24×24)	SKLN12 (28×28)
Cooling Volume (m³)	2.5	4	6	8	12
Capacity (t/h)	6	10	15	20	30
Cooling Time (min)	Not less than 10-15				
Product Temp.after Cooling	Not higher than ambient temperature 3-5°C				
Air Absorbing Volume (m³/min.t)	34				
Power (kw)	1.5	2.2	2.2	2.2	1.5+2.2
Power of airlock (kw)	0.75	1.1	1.1	1.5	1.5

FEED PELLET CRUMBLER

KEY FEATURES

- Using four rolls to form two sets of grinding zones, the output is doubled.
- Compared with twin-roll series with the same capacity, significantly shorten the length of the rolls.
- Manufacturing accuracy of the rolls is high, even distribution on the roll, the efficiency is high.
- Easy maintenance and easy installation.



Model	Power (kW)	Capacity (t/h)
SSLG20×140×4	11×2+1.1×2	2~5
SSLG25×170×4	18.5×2+1.1×2	4~9
SSLG30×180×4	22×2+1.1×2	8~12

DRYER MACHINE

It is suitable for the drying treatment of puffed feed in the feed industry. It is also suitable for the drying treatment of powder, flake, and strip materials in feed, food and chemical industries.



Model	Circulating Fan Power(kw)	Middle Section Number	Net belt drying area (m ²)	Capacity (t/h)
SHGW24x2-6	7.5x8	4	48	4.5-5.5
SHGW24x2-7	7.5x10	5	58	6-7
SHGW24x2-9	7.5x12	7	68	8-10
SHGW24x2-10	7.5x16	8	88	10-12

VIBRATION SCREENER

KEY FEATURES

- Vibration screener adopts the principle of vibration screening, with reasonable screen angle (adjustable) and hole.
- Chain is adopted to clean screen to strengthen the screen and guarantee the screen effect.
- Compact structure, stable working and convenient operation & maintenance.
- The vibration screener's vibration motor is with small vibration, low noise and smooth performance.



Model		Power(kw)	Capacity (t/h)	
			Mash Feed	Pellet Feed
2C Series	SFJH 80x2C	1.5	4-6	5-8
	SFJH110x2C	2.2	6-8	8-10
	SFJH130x2C	4	8-13	10-15
	SFJH140x2C	5.5	12-16	14-18
	SFJH150x2C	5.5	16-18	15-20
	SFJH165x2C	7.5	18-20	18-25

Maize Pre-Cleaning Unit (CP Series)

Remove coarse impurities, fines, dust, stones and ferrous particles from raw maize before drying /milling/storage—reduces wear on downstream equipment and improves storage life.

- Fast tool-less screen change with side access doors
- Adjustable deck inclination and throw for different grains
- Transparent inspection windows & LED service light (option)
- Low vibration base frame with isolation mounts
- Plug-and-run control panel with VFD for fan (option)
- Compact footprint; easy ducting to existing dust system



COUNTER FLOW FEED PELLET CLEANER



DRUM CLEANER



PACKING SCALES



SDBLY-PD belt
Packaging scale



Loading belt
Conveyer





BUCKET ELEVATOR

It can be used for the vertical transportation of pellets, powder and bulk materials. It is widely used in feed mill plants, food plants, chemical plants, construction materials, grain storage and ports.

Strong construction and high anti-wear components make maintenance easy and extend service life.



Chain Conveyor

CAPACITIES FROM : 15 – 280 m³/h



Screw Conveyor

CAPACITIES FROM : 5 – 120 m³/h



Trough Conveyor

CAPACITIES FROM : 15 – 475 M³ Per Hour



FEED LINE ROTARY DISTRIBUTOR



Oil Adding Machine



Automatic Liquid Adding Machine

ANCILLARY EQUIPMENT



Pneumatic
Gravity Diverter



Motorized
Gravity Diverter



Double Wing
Gate



Pneumatic
Slide Gate



Manual Slide
Gate



Motorized Slide
Gate

ANCILLARY EQUIPMENT



High Pressure Fan



Blower Fan



Air Fan



Centrifugal Fan



Air Lock



Rotatory Feeder